

Munkres 7

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1 Introduction

There exists a metric $\bar{\rho}$ for which $(\mathbb{R}^\omega, \bar{\rho})$ is complete. Let $\bar{\rho}(x, y) = \sup_{n < \omega} \rho_n(x_n, y_n)$, where $\rho_n(x_n, y_n) = \min(1, |x_n - y_n|)/n$.

Suppose $U(x) \subset \mathbb{R}^\omega$ is a basis element. Let $K \subset N$ be the finite set of n for which $U_n \neq \mathbb{R}$. For all $n \in K$, pick $B(x_n, n\epsilon_n) \subset U_n$ and $n\epsilon_n < 1$. Let $\epsilon = \min_{n \in K} \epsilon_n > 0$.

Suppose $y \in B_{\bar{\rho}}(x, \epsilon)$. Take $n \in K$ (since $y_n \in U_n$ otherwise). Then $\rho_n(x_n, y_n) < \epsilon_n \rightarrow \min(1, |x_n - y_n|) < n\epsilon_n$. Since $n\epsilon_n < 1$, we have $|x_n - y_n| < n\epsilon_n \rightarrow y_n \in B(x_n, n\epsilon_n) \subset U_n$. Hence, $B_{\bar{\rho}}(x, \epsilon) \subset U$.

Suppose $B_{\bar{\rho}}(x, \epsilon)$ is a basis element (we can assume centered around x). Pick N such that $1/N < \epsilon$. For all $n \geq N$, $\rho_n(x_n, y_n) \leq 1/n < \epsilon$, so take $U_n = \mathbb{R}$. For $n < N$, take $U_n = B(x_n, \epsilon)$. If $y \in U$, then for $n < N$, $\rho_n(y, x) = \min(1, |y_n - x_n|)/n < \epsilon/n \leq \epsilon \rightarrow \rho_n(y, x) < \epsilon$. For $n \geq N$, $\rho_n(y, x) = \min(1, |y_n - x_n|)/n \leq 1/n < \epsilon$, so $\bar{\rho}(x, y) < \epsilon$, ie $U \subset B_{\bar{\rho}}(x, \epsilon)$.

Let $m \in N$ be fixed, $0 < \epsilon < 1$. Pick N such that $i, j \geq N \rightarrow \bar{\rho}(x_i, x_j) < \epsilon/m$. Then $\rho_m(x_{im}, x_{jm}) < \epsilon/m \rightarrow \min(1, |x_{im} - x_{jm}|) < \epsilon \rightarrow |x_{im} - x_{jm}| < \epsilon$, so (x_{nm}) is Cauchy for fixed m .

By completeness of $\mathbb{R}$, let $x_{nm} \rightarrow x_{\infty m}$. Then (x_n) is a sequence with each point $x_{nm} \rightarrow x_{\infty m}$, so $x_n \rightarrow (x_{\infty m})_{m \in N}$ in the product topology, so $(\mathbb{R}^\omega, \bar{\rho})$ is complete.

Let X be first-countable. Then X is compactly generated. Let $A \subset X$ have the property for all compact $C \subset X$, $A \cap C$ is open in C .

Let $x \in A$ and $\mathcal{N}_x = \{N_1, N_2, \dots\}$ be a neighborhood basis of x such that $N_1 \supset N_2 \supset N_3 \supset \dots$. Suppose FTSOC $N_i \not\subset A$ for all $i \in N$. Pick $x_i \in N_i - A$ and let $X' = \{x_1, x_2, x_3, \dots\} \cup \{x\}$, which I claim is compact.

Let $\mathcal{A}$ be an open covering of X' . Pick $U(x) \in \mathcal{A}$ and $n_0 \in N$ such that $N_{n_0} \subset U$.

Then for all $n \geq n_0$, $x_n \in N_n \subset N_{n_0} \subset U$. The remainder $x_1, x_2, x_3, \dots, x_{n_0-1}$ can be finitely covered.

By assumption, $\{x\} = X' \cap A = X' \cap V$ for some V open in X . Pick $n \in N$ such that $N_n \subset V$. Then $x_n \in V \cap X'$, cont.

2 Homotopy

Let $f : X \times Z \rightarrow Y$ be continuous and $\mathcal{C}(X, Y)$ have the compact-open topology. Then $F : Z \rightarrow \mathcal{C}(X, Y)$ is continuous, where $F(z)(x) = f(x, z)$ (fix $z \in Z$, vary $x \in X$). If X is locally compact Hausdorff, the converse holds.

Let $F(z_0) \in \bigcap S(C, U)$ be a basis element of $\mathcal{C}(X, Y)$. First consider $F(z_0) \in S(C, U)$. Since f is continuous, for all $x \in C$, pick $(W \times V)(x, z_0)$ such that $f(W \times V) \subset U$. By compactness of C , pick a finite subcover $\{W_n\}$. Let $V = \bigcap V_n$.

I claim $F(V) \subset S(C, U)$. Let $F(z) \in F(V)$, $x \in C$, so $x \in W_n$ for some n . Since $z \in V \subset V_n$, $F(z)(x) = f(x, z) \in U$, so $F(z)(C) \subset U \rightarrow F(z) \in S(C, U)$.

Since the intersection $\bigcap S(C, U)$ is finite, we may intersect finitely many V 's so that $F(V) \subset \bigcap S(C, U)$.

Conversely, suppose $F : Z \rightarrow \mathcal{C}(X, Y)$ is continuous, X is locally compact Hausdorff. Let $(x_0, z_0) \in X \times Z$, $U(F(z_0)(x_0)) \subset Y$ be open. Since $F(z_0)$ is continuous and X is locally compact Hausdorff, pick open $W(x_0)$ such that $\overline{W}$ is compact and $F(z_0)(\overline{W}) \subset U$. Since $S(\overline{W}, U)$ is open in $\mathcal{C}(X, Y)$ and F is continuous, pick $V(z_0)$ such that $F(V) \subset S(\overline{W}, U)$.

I claim $f(W \times V) \subset U$. Let $(x, z) \in W \times V$. Since $F(z) \in F(V) \subset S(\overline{W}, U)$, $F(z)(W) \subset F(z)(\overline{W}) \subset U$, or $f(W, z) \subset U$. Varying $z \in V$ gives $f(W \times V) \subset U$.

3 Function Spaces Again

If Y is Hausdorff, so is $\mathcal{C}(X, Y)$. Let $f \neq g \in \mathcal{C}(X, Y)$. Then $f(x) \neq g(x)$ for some $x \in X$. Since Y is Hausdorff, pick $U(f(x)), V(g(x))$ disjoint, so $f \in S(\{x\}, U), g \in S(\{x\}, V)$ open disjoint.